A Project Report

on

DIGITAL BRAIN TWIN FOR PERSONALIZED NEUROLOGY

carried out as part of the ***Project Based Learning-1 (DSE2170)***

Submitted

by

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in partial fulfilment for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering (Data Science)

School of Computer Science and Engineering
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MANIPAL UNIVERSITY JAIPUR, JAIPUR
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November 2025

CERTIFICATE

This is to certify that the project-based learning , project titled Digital Brain Twins for Personalized Neurology is a record of the bonafide work done by **Anusha Dayal** (2430010333), **Namrata Kothari** (2430010345) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Data Science) of Manipal University Jaipur, Jaipur during the academic year 2025-26.

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ABSTRACT

The prevalence and complexity of neurological disorders require a paradigm shift toward personalized therapeutic strategies. Clinical approaches to date often rely on generalized trials, animal models, and invasive diagnostics, leading to suboptimal, delayed, and patient-nonspecific outcomes. This project meets this critical need by proposing the development of patient-specific, generative, and adaptive computational models called Digital Brain Twins. The key aim is to develop a non-invasive, multi-scale framework capable of integrating diverse, individual-patient data (e.g., MRI, EEG/MEG, and clinical records) in order to replicate the neurobiological state of the subject with detail.

It will serve as an in-silico testing platform for the predictive simulation of clinical interventions, ranging from targeted drug responses to presurgical planning. Such a model supports clinical decision-making through high-fidelity, patient-specific predictions by bridging microscopic biological mechanisms, such as synaptic degeneration, with macroscopic network dynamics. Deployment of this framework will effectively minimize patient risk, treatment delays, and accelerate the realization of precision medicine in neurology.

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1. INTRODUCTION

1.1Background

Personalized medicine is transforming healthcare; however, treatments for complex neurological disorders, such as Epilepsy and Parkinson's Disease, remain heavily generalized in clinical paradigms and trial-and-error approaches. Such methodologies frequently lead to substantial delays in treatment and suboptimal patient outcomes, given the lack of specificity concerning the individual's unique neurobiological state. This context emphasizes the pressing need for non-invasive, high-fidelity predictive tools.

1.2 Purpose of the Document

This document aims to present a formal research proposal for the development of a DBT framework. It recognizes a deficiency in contemporary clinical neurology, delineates a comprehensive multi-scale computational technique, defines explicit project objectives and deliverables, and articulates the validation strategy via proposed clinical case studies.

1.3 Scope of the Project

The project is focused on the development of a virtual patient-specific computational model of the brain that can simulate disease dynamics and the response to treatments. The main emphasis has been placed on multi-scale computational modelling together with AI/ML parameterization based on existing non-invasive clinical data, such as MRI, EEG/MEG, and records for personalized neurology. Clinical focuses will include predictive simulation of treatment outcomes for Epilepsy and Parkinson's Disease.

2. LITERATURE REVIEW

2.1 Overview of Existing Solutions

Current approaches to computational brain modelling exist on two extremes: highly detailed, localized single-cell models and lower-resolution large-scale whole-brain network models (connectome-based models). Current personalized whole-brain models, such as those for Epilepsy and AD, integrate both structural and functional connectivity data to map activity and estimate parameters in a subject-specific brain space.

Comparison of Existing Computational Brain Modeling Techniques

Feature	Single-Cell Models (Microscale)	Whole-Brain Network Models (Macroscale)	Digital Brain Twin (DBT) (Multi-Scale)
Scale	Subcellular to Cellular (Micro/Meso)	Global Network Dynamics (Macro)	Full Multi-Scale Integration (Micro to Macro)
Primary Data Focus	Ion channels, Synapses, Neurons	Structural/Functional Connectivity (Connectome)	Multi-modal (MRI, EEG/MEG, Clinical Records)
Main Goal	Detailed physiological mechanism	Mapping activity, connectivity patterns	Predictive Simulation, Personalized Treatment
Key Limitation	Not scalable to the whole brain	Lacks biological mechanism/detail	Complexity, Data Integration/Validation Challenge

2.2 Identified Challenges

A key challenge involves basing diagnoses and treatments on general clinical paradigms, which results in outcomes not being optimized for the individual patient's unique biological mechanisms.Moreover, traditional models fail to connect microscopic phenomena-such as synaptic degeneration or ion channel properties-with macroscopic clinical observations like seizure severity or cognitive decline. This multilevel link is actually the fundamental scientific and engineering gap.

2.3 Areas of Improvement

There is a recognized opportunity to develop a unified framework that integrates multiple biological scales, from molecular/cellular processes to whole-brain networks, allowing for a comprehensive mechanistic understanding of disease. The DBT frameworks allow for better patient stratification and functional insight into the processes involved in disease progression across various disorders.

2.4 Summary of Findings

The literature confirms that personalized modelling is possible but inherently requires a multi-scale, generative, and adaptive DBT framework that captures the integration of diverse data types with a mathematical connection between micro-pathological causes and macro-network effects. Only then can such modelling move beyond descriptive modelling toward predictive simulation.

3. PROBLEM DEFINITION

3.1 Project Scope

The development of a computational framework, the design and development of the multi-scale coupling algorithms, and the AI/ML pipeline for personalized parameter tuning define the scope of this project. The model will be validated using existing retrospectively collected clinical data, focusing particularly on pre-surgical and pharmacological outcomes prediction.

3.2 Objectives

The main goals of this project are:

1. Design and implement a patient-specific, multi-scale computational model able to represent individual brain structure, function, and the underlying biological processes.
2. To build an AI/ML pipeline for the integration of multi-modal patient data robustly, namely, MRI and EEG/MEG, to personalize and validate model parameters.
3. To use the DBT as an in-silico platform to predict the efficacy of surgical and pharmacological interventions in key neurological disorders.

3.3 Deliverables

Key deliverables of the project include:

1. DBT Model Framework : An overall documented computational engine, including multi-scale coupling logic.
2. AI/ML Parameterization Pipeline: Code and workflow for automated model tuning and validation against patient-specific data.
3. Case Study Simulation Results: Predictive output and analysis of pre-surgical planning (Epilepsy) and pharmacological response (Parkinson's).

3.4 Problem Statement

Current neurological treatments are limited by a reliance on generalized, non-specific approaches that often yield suboptimal outcomes with considerable patient risk. This is fundamentally due to the absence of a high-fidelity, non-invasive, and patient-specific tool that can accurately simulate the neurological response to therapeutic interventions prior to clinical application.

4. PROPOSAL DEVELOPMENT

4.1 Research Findings

This proposal is driven by one key finding, namely that personalized brain models have demonstrated the ability to link progression of cognitive decline with underlying synaptic and connectivity degeneration. However, true predictive simulation requires the robust integration of these multi-scale links, hence the need for a formal DBT approach.

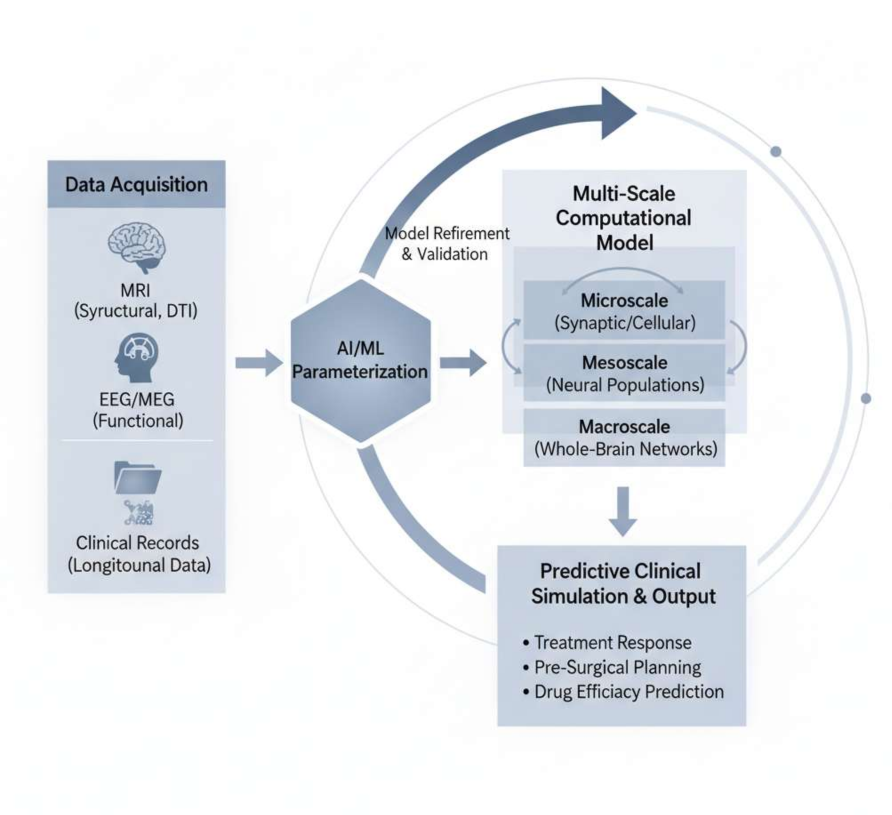
4.2 Project Objectives

As indicated in Section 3.2, the main objective of the project is to develop a predictive multi-scale Digital Brain Twin framework for personalized neurological treatment simulation.

4.3 Methodology

The methodology combines advanced data integration, multi-scale mathematical modeling, and machine learning parameterization in order to achieve the project's objectives, this is conceptually illustrated in the figure below.

Figure 1: Concepual Architecture the Digital Brain Twin (DBT) Framework



4.3.1 Data Collection Methods

Data sources encompass:

• MRI: Magnetic Resonance Imaging (MRI) is employed for structural data, alongside Diffusion Tensor Imaging (DTI), which delineates the brain's structural connectome.

• EEG/MEG: to record patterns of functional activity and network behavior that change over time.

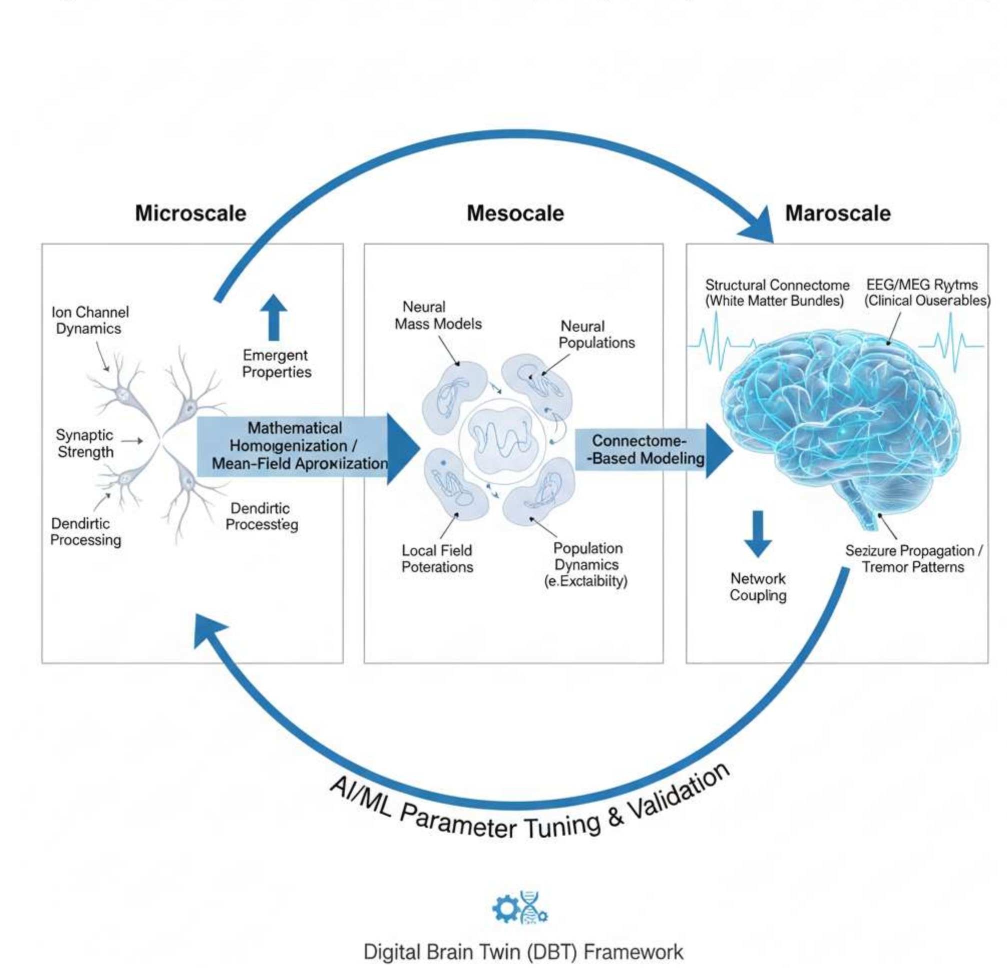
• Clinical Records: Longitudinal data about patient symptoms, drug history, and clinical outcomes for parameter fitting and validation.

4.3.2 Analytical Techniques

Multi-Scale Model Architecture:

We define such architecture operating at multiple scales, mathematically linking mesoscale/microscale pathological mechanisms with the resulting macroscale effects on whole-brain network dynamics and functional connectivity. This involves using mean-field modelling to represent large populations of neurons coupled at the mesoscale by the structural connectivity derived from the patient's MRI at the macroscale.

Figure 2: Schematic of Multi-Scale Modeling Integration (Microscale to Macroscale)



AI/ML Parameterization and Validation:

Two critical functions will utilize the implementation of an AI/Machine Learning Pipeline.

1. Parameter Estimation: Adjusting differential equations and control parameters of the multi-scale model, including neuronal excitability and connection strengths, to align with the patient's empirical data, such as power spectra derived from EEG.

2. Comprehensive Validation: The simulation of anticipated patient trajectories such as post-surgical functionality and drug-induced alterations compared to real longitudinal clinical outcomes.

4.4 Outcomes

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The successful development of DBT will therefore offer an internationally leading predictive simulation platform to eventually enhance the effectiveness and personalization of neurological care. The aim is to reduce patient risk in procedures and accelerate therapeutic innovation, as clinical treatment planning moves from trial-and-error practices to data-driven simulation.

4.5 Timeline and Milestones

Phase	**Milestone**	**Planned Completion**
Phase 1: Topic Selection	The first stage is to get approval for the project title, scope, and other details.	9th October 2025
Phase 2: PRS-1	The first Project Report, which includes the problem statement, an assessment of the literature, and the suggested technique, has been turned in.	15th October 2025
Phase 3: Internal assessment	A formal presentation and viva examination of the project's foundational work and progress was done.	4th November
Phase 4: End term assessment	The final evaluation, including submission of the complete project report and demonstration of all deliverables.	13th November

5. CONCLUSION

5.1 Summary

The paper describes the design of a patient-specific, multiscale computational model called the Digital Brain Twin (DBT), which will overcome significant limitations in generalized neurological treatments. Combining multi-modal patient data using a sophisticated AI/ML pipeline, the DBT has the potential to be a non-invasive high-fidelity platform for predictive simulation of clinical interventions targeted at pre-surgical planning and pharmacological response.

5.2 Next Steps

Next steps will be to finalize which mathematical models-specifically, neural mass models-will be utilized at the mesoscale, outline a data normalization and integration strategy in detail, and begin implementing the core data processing pipeline according to the timeline outlined in Phase 2. Future work will also focus on addressing the regulatory and ethical considerations for clinical adoption.

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